# Function extensionality from the interval type: pointwise equal functions are equal

The Univalent Foundations Program

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*Editorial note.* The counted statement is the author's 'thm:interval-funext' (source lines 404-408, Lemma 6.3.2); the proof environment begins after one blank line, so the pairing is by adjacency.

*Editorial note.* The retained context is the author's own lead-in paragraph, which is needed because it records that the function extensionality axiom is suspended during this proof.

*Editorial note.* The retained counter values (chapter 6, section 3, theorem offset 1) make the retained lemma print as Lemma 6.3.2, matching the original book numbering.

*Editorial note.* The single external reference points at another chapter's section that the retained context invokes by label; it is a navigation pointer only.

*Editorial scope.* One original result of the HoTT Book, counted once: function extensionality from the interval type (Lemma 6.3.2).

*Allowed standard prerequisites (editorial).*

- The interval as a higher inductive type. The interval I is generated by $0 : I$, $1 : I$ and $\mathrm{seg} : 0 = 1$, with an induction principle and a recursion principle whose computation rules are definitional at the three constructors.. The proof builds maps I -> B and I -> (A -> B) by the recursion principle with prescribed values at 0, 1 and seg (source lines 411-417 and 418-419).
- Function extensionality may not be assumed here. For the duration of the proof the function extensionality axiom is suspended, so f = g in A -> B must be produced by an actual path.. The conclusion at source line 422 is the path ap q (seg) in the function type, not an application of any extensionality axiom; the retained context supplies the suspension.

*Source reference.* Original section 4.9: sources/hott-hits/book-labelnumbers-first-edition.tex, source lines 1718–1718. Pointer to the earlier section where the same suspension of the function extensionality axiom is announced. It is a navigation pointer only: the retained proof of the counted lemma is complete and does not use that section's argument.

However, type-theoretically the interval does still have some interesting features, just like the topological interval in classical homotopy theory. For instance, it enables us to give an easy proof of function extensionality. (Of course,

as in original section 4.9, for the duration of the following proof we suspend our overall assumption of the function extensionality axiom.)

**Lemma 6.3.2.** *If $f, g : A \to B$ are two functions such that $f(x) = g(x)$ for every $x : A$, then $f = g$ in the type $A \to B$.*

*Proof.* Let's call the proof we have $p : \prod_{(x:A)}(f(x) = g(x))$. For all $x : A$ we define a function $\widetilde{p}_x : I \to B$ by

$$\begin{aligned} \widetilde{p}_x(0_I) &:\equiv f(x), \\ \widetilde{p}_x(1_I) &:\equiv g(x), \\ \widetilde{p}_x(\mathsf{seg}) &:= p(x). \end{aligned}$$

We now define $q : I \to (A \to B)$ by

$$q(i) :\equiv (\lambda x.\, \widetilde{p}_x(i))$$

Then $q(0_I)$ is the function $\lambda x.\, \widetilde{p}_x(0_I)$, which is equal to $f$ because $\widetilde{p}_x(0_I)$ is defined by $f(x)$. Similarly, we have $q(1_I) = g$, and hence

$$q(\mathsf{seg}) : f =_{(A \to B)} g$$

□